WHEN I WAS A CHILD, my most exultant dreams were about flying—not in some machine, but all by myself. I would be skipping or hopping, and slowly I could pull my trajectory higher. It would take longer to fall back to the ground. Soon I would be on such a high arc that I wouldn't come down at all. I would alight like a gargoyle in a niche near the pinnacle of a skyscraper, or gently settle down on a cloud. In the dream—which I must have had in its many variations at least a hundred times—achieving flight required a certain cast of mind. It's impossible to describe it in words, but I can remember what it was like to this day. You did something inside your head and at the pit of your stomach, and then you could lift yourself up by an effort of will alone, your limbs hanging limply. Off you'd soar.

I know many people have had similar dreams. Maybe most people. Maybe everyone. Perhaps it goes back 10 million years or more, when our ancestors were gracefully flinging themselves from branch to branch in the primeval forest. A wish to soar like the birds motivated many of the pioneers of flight, including Leonardo da Vinci and the Wright brothers. Maybe that's part of the appeal of spaceflight, too.

In orbit about any world, or in interplanetary flight, you are literally weightless. You can propel yourself to the spacecraft ceiling with a slight push off the floor. You can go tumbling through the air down the long axis of the spacecraft. Humans experience weightlessness as joy; this has been reported by almost every astronaut and cosmonaut. But because spacecraft are still so small, and because space "walks" have been done with extreme caution, no human has yet enjoyed this wonder and glory: propelling yourself by an almost imperceptible push, with no machinery driving you, untethered, high up into the sky, into the blackness of interplanetary space. You become a living satellite of the Earth, or a human planet of the Sun.

Planetary exploration satisfies our inclination for great enterprises and wanderings and quests that has been with us since our days as hunters and gatherers on the East African savannahs a million years ago. By chance—it is possible, I say, to imagine many skeins of